Training data

$$\begin{array}{c} \quad \underline{f_1} \quad \underline{f_2} \quad \underline{f_3} \qquad\qquad \underline{f_{14}} \\ X_{train} = X = \begin{pmatrix} & & & \\ & 339 \times 14 & & \\ & & & \end{pmatrix} \\ \quad \text{CRIM} \quad \text{ZON} \quad \text{INDUS} \end{array} \qquad \begin{array}{c} \underline{y} \\ \begin{pmatrix} \\ \\ \\ \end{pmatrix} \end{array} \; 339 \times 1.$$

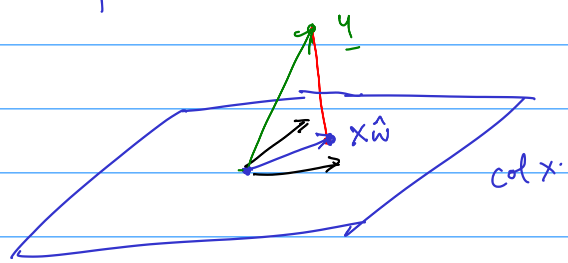
Set of all linear combinations of columns of X : $\text{col}(X)$.

$X\underline{w}$: If $\underline{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_{14} \end{pmatrix}$ $\quad X\underline{w} = w_1\underline{f_1} + w_2\underline{f_2} + \dots + w_{14}\underline{f_{14}}$

Attempt 0   Maybe $\underline{y}$ can be written as $X\underline{w}$?

Solve $\underline{y} = X\underline{w}$. $[X \; \underline{y}]$

Most likely there is no solution.



$\underline{y}$

$X\underline{\hat{w}}$

$\underline{y} - X\underline{\hat{w}} := \underline{e}$

Red line orthogonal to plane: Red line is $\perp$ to every vector in plane.

Suppose $M = \{\underline{v}_1 \dots \underline{v}_n\}$ is a maximal linearly indpt set in $\text{col}(X)$.

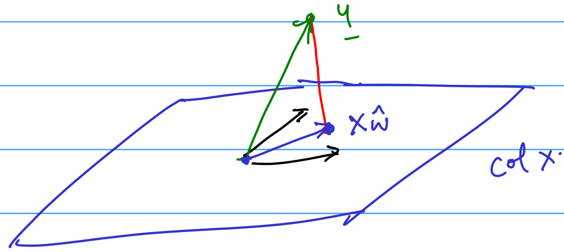
(What would be a maximal lin. ind. set in $\text{col}(X)$?)

Suppose. $\underline{e}^T\underline{v}_1 = \underline{e}^T\underline{v}_2 \dots \underline{e}^T\underline{v}_n = 0$.

Pick $\underline{x} \in \text{col}(X)$. Now $[\underline{v}_1 \dots \underline{v}_n \; \underline{x}]$ has the last col free.

$\therefore \underline{x} = \begin{bmatrix} \underline{v}_1 & \dots & \underline{v}_n \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$.

$$\underline{e}^T\underline{x} = \underline{e}^T \begin{bmatrix} \underline{v}_1 & \dots & \underline{v}_n \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = \begin{bmatrix} \underline{e}^T\underline{v}_1 & \dots & \underline{e}^T\underline{v}_n \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = 0.$$



$\underline{y}$

$X\hat{\underline{w}}$

$\underline{y} - X\hat{\underline{w}} = \underline{e}$

What is the closest point to y? It's $X\hat{w}$ such that $y - X\hat{\underline{w}}$ is orthogonal to every pivot col of X.

Assuming all cols of X are pivot cols. $y - X\hat{\underline{w}} \perp$ every col of X.

$$X = \begin{bmatrix} | & & | \\ \underline{f}_1 & \cdots & \underline{f}_{14} \\ | & & | \end{bmatrix} \qquad X^T = \begin{bmatrix} — & \underline{f}_1^T & — \\ & \vdots & \\ — & \underline{f}_{14}^T & — \end{bmatrix}$$

$$X^T \underline{e} = \begin{bmatrix} — & \underline{f}_1^T & — \\ & \vdots & \\ — & \underline{f}_{14}^T & — \end{bmatrix} \underline{e} = \begin{bmatrix} \underline{f}_1^T \underline{e} \\ \vdots \\ \underline{f}_{14}^T \underline{e} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \underline{0}.$$

$X^T \underline{e} = \underline{0}$ is the formalization that $\underline{e}$ is $\perp$ to all vectors in Col(X).

$$\underline{0} = X^T \underline{e} = X^T (\underline{y} - X\hat{\underline{w}}).$$
$$= X^T \underline{y} - X^T (X\hat{\underline{w}})$$
$$= X^T \underline{y} - (X^T X)\hat{\underline{w}}$$

$X : n \times p$. $\qquad X^T : p \times n$

$\underset{p\times n}{X^T}\,\underset{n \times p}{X} : p \times p.$ $\qquad (X^T X)^T = X^T (X^T)^T = \underline{X^T X}$

$$(AB)^T = B^T A^T$$

$$\left[\begin{pmatrix} | & | \\ \underline{a}_1 & \underline{a}_2 \\ | & | \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}\right]^T = \left[b_1 \underline{a}_1 + b_2 \underline{a}_2 \right]^T = b_1 \underline{a}_1^T + b_2 \underline{a}_2^T$$
$$= (b_1 \quad b_2) \begin{bmatrix} — & \underline{a}_1^T & — \\ — & \underline{a}_2^T & — \end{bmatrix}.$$
$$= \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}^T A^T.$$

$$\begin{pmatrix} a & c \\ d & b \end{pmatrix} \qquad \text{Transpose} \qquad \begin{pmatrix} a & d \\ c & b \end{pmatrix}$$

Therefore

$$\underline{0} = X^T \underline{e} = X^T(\underline{y} - X\hat{\underline{w}})$$
$$= X^T \underline{y} - X^T(X\hat{\underline{w}})$$
$$= X^T \underline{y} - (X^T X)\hat{\underline{w}}$$

$$(X^T X)\hat{\underline{w}} = X^T \underline{y}$$

vector of coefficients for our proposed linear model.

$X^T X$: How many pivot cols?

X : $n \times p$. $\quad X^T X$: $p \times p$.

Suppose c3 of X = c1 of X + c2 of X.

$$X^T X \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = X^T \underline{0} = \underline{0}$$

c4 of X = c1 of X − c2 of X.

$$X^T X \begin{pmatrix} 1 \\ -1 \\ 0 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = X^T \underline{0} = \underline{0}.$$

Suppose $\underline{z} \in \text{null}(X)$ $\quad X\underline{z} = \underline{0}$

then $X^T X \underline{z} = X^T \underline{0} = \underline{0}$ $\quad \underline{z} \in \text{null}(X^T X)$.

$$\text{null}(X) \subseteq \text{null}(X^T X) \qquad (1)$$

Suppose $X^T X \underline{v} = \underline{0}$.

then $\underline{v}^T X^T X \underline{v} = \underline{v}^T \underline{0} = 0$

$X\underline{v}$ is a vector $\quad (X\underline{v})^T = \underline{v}^T X^T$.

$\therefore (X\underline{v})^T X\underline{v} = 0$. $\quad$ But for any vector $\underline{u}$, $\underline{u}^T \underline{u} = \|\underline{u}\|^2$.

$$\|X\underline{v}\|^2 = 0 \quad \therefore \quad X\underline{v} = 0. \quad \therefore \ \text{null}(X^TX) \subseteq \text{null}(X) \quad \text{— (2)}$$

$$\therefore \ \text{null}(X) = \text{null}(X^TX)$$

In particular if X has pivot in each col, X^TX has too.

$$\underset{p \times p}{(X^TX)}\,\hat{\underline{w}} = X^T\underline{y}. \qquad \hat{\underline{w}} = (X^TX)^{-1}X^T\underline{y}.$$

$$\left[\ \underbrace{X^TX}_{p \text{ cols}} \qquad X^T\underline{y}\ \right].$$

p rows.

To compute inverse of A. $(n \times n)$

$$BA = I$$

(collect all row operations to bring A to its rref)

We saw if $BA = I$ then $AB = I$.

Solve

$$A\underline{x}_1 = \begin{pmatrix}1\\0\\0\\ \vdots \\0\end{pmatrix} \qquad \left[A \ \begin{matrix}1\\0\\0\\0\\0\end{matrix}\right] \qquad [A, \ I]$$

$$A\underline{x}_2 = \begin{pmatrix}0\\1\\0\\ \vdots \\0\end{pmatrix} \qquad \left[A \ \begin{matrix}0\\1\\0\\0\\0\end{matrix}\right]. \qquad \downarrow$$

$$[I \quad A^{-1}].$$

$$\vdots$$

$$A\underline{x}_n = \begin{pmatrix}0\\ \vdots \\0\\1\end{pmatrix}$$

If $A\underline{x} = \begin{pmatrix}1\\1\\1\end{pmatrix}$ $\quad A\underline{x} = \begin{pmatrix}1\\1\\0\end{pmatrix}$ $\quad A\underline{x} = \begin{pmatrix}1\\0\\0\end{pmatrix}.$ all have solutions.

Key: A has a pivot in each of its three rows.

* A has 3 pivots.
* If A is square all solutions above are unique.
* If A is square it has an inverse.
* $A\underline{x} = \underline{b}$ has at least one soln for all $\underline{b}$.

$$\begin{pmatrix}1 & 1 & 1\\ 1 & 1 & 0\\ 1 & 0 & 0\end{pmatrix} \qquad \left(\begin{array}{ccc|c}1 & 1 & 1 & \underline{b}\\ 0 & 1 & 1 & \\ 0 & 0 & 1 & \end{array}\right).$$

$$\left(A \ \middle| \ \begin{matrix}1 & 1 & 1\\ 1 & 1 & 0\\ 1 & 0 & 0\end{matrix} \ \middle| \ \begin{matrix}\underline{b}\\ \\ \end{matrix}\right).$$

A has three rows

Suppose $\underline{x}$ is unique soln of $A\underline{x} = \underline{b}$ for a square A and fixed $\underline{b}$

* $A\underline{x} = \underline{0}$ has non-zero solutions. False.
* A has no inverse. False.
* there is at least one $\underline{z}$ s.t. $A\underline{x} = \underline{z}$ has no soln. False. ←
* the rref of $A = I$. True.
* $A\underline{x} = \underline{c}$ has exactly one soln for all $\underline{c}$. True. ←

$$\begin{bmatrix} A & \underline{z} \end{bmatrix}.$$

* If a vector $\underline{v}$ is orthogonal to every row of a matrix A.

$A\underline{v} = \underline{0}$.

⇒ another way of saying $\underline{v} \in null(A)$.

* How to get null space of A?

elimination on A → rref (A)

$$rref(A) = \begin{array}{ccccc} c1 & c2 & c3 & c4 & c5 \\ \end{array}$$
$$\begin{pmatrix} 1 & 0 & -1 & 0 & 2 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

$\begin{array}{ccccc} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ \text{pivot} & \text{pivot} & \text{free} & \text{pivot} & \text{free} \end{array}$.

$c3 = -c1 + c2$.

$c3 + c1 - c2 = 0$.

$$A \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \underline{0}.$$

$c5 = 2c1 + c4$. $c5 - 2c1 - c4 = 0$.

$$A \begin{pmatrix} -2 \\ 0 \\ 0 \\ -1 \\ 1 \end{pmatrix} = \underline{0}.$$